Md Zafar Iqbal, PhD.

Postdoctoral Associate, Department of Agricultural and Biological Engineering, University of Florida

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Education

PhD in Biological & Agricultural Engineering from Texas A&M University, USA. Dec 2024

Dissertation title: Round Cotton Modules Monitoring System: A Smart Solution for Efficient Module Management Between Harvest and Ginning.
Committee: Robert G. Hardin, Azlan Zahid, John R.C. Robinson, Joshoua Peeples, Edward M. Barnes

MSc in Bioproduction Machinery Engineering from Chungnam National University, South Korea. Aug 2019

Thesis title: Design of a Gear Driven Hopper Type Dibbling Mechanism for a 2.7 kW Two-row Pepper Transplanter.
Committee: Sun-Ok Chung, Yong-Joo Kim, Dae-Hyun Lee

BSc in Agricultural Engineering from Hajee M. Danesh Science & Technology University, Bangladesh. May 2017

Thesis title: Development of a Low-cost Smart Evaporative Cooling System for Storing Fresh Tomatoes.
Committee: Md Shaha Nur Kabir, Muhammad Abdul Munnaf

Experiences

- **Postdoctoral Research Associate**

  Employed at the University of Florida since Fall 2024 within University of Florida's Agricultural Robotics and Mechatronics research group led by Dr. Tom Burks. Research focuses on developing artificial intelligence and computer vision solutions for precision agriculture and food safety through multimodal learning, deep learning, and intelligent sensing. Areas of application include plant disease diagnosis in citrus, soybean, grape, and spinach, morphological phenotyping, and detection of Escherichia coli contamination on plant surfaces and food-processing materials using non-destructive imaging. Core expertise includes deep learning, multimodal data fusion, vision transformers (ViTs), convolutional neural networks (CNNs), hyperspectral image analysis, RGB-D imaging, machine vision, image processing, and explainable AI. Developed hybrid AI architectures integrating transformer-based global feature learning with convolutional spectral feature extraction to improve classification accuracy, robustness, and computational efficiency. Planned research over the coming months will also involve integrating robotic platforms, including autonomous mobile robots (Clearpath Ridgeback, Rover MiTi, Rover Mega) and robotic manipulators (Kinova), for indoor and outdoor crop phenotyping, disease inspection, and automated data collection. Additional contributions include customization of hyperspectral imaging systems, development of RGB-D sensing platforms, design of graphical user interfaces (GUIs) for automated data acquisition and analysis, and development of smartphone-deployable AI systems for real-time plant disease diagnosis and geospatial disease mapping on UAV and UGV platforms.

- **Research Experiences**

  Served as **Graduate Research Assistant** (Spring 2020-Fall 2023), upgraded and tested an AI-enabled monitoring system for high-quality cotton production by integrating multimodal sensing data, including moisture, temperature, GPS routing, image data, and RFID-based module tracking. Implemented deep learning-based computer vision models, including CNNs, for automated detection of cotton modules with damaged covers, enabling real-time quality assessment, improved operational efficiency, and enhanced production management.

  Previously served as **Graduate Research Assistant** at Chungnam National University (Fall 2017-Fall 2019), focused on the design and development of a robotic picking and dibbling mechanism for a walking-type pepper transplanter, along with operational overturning and service life analyses. Also developed a remote environmental and pest monitoring and control system for smart greenhouses using WSN and IoT technologies.

  Undergraduate research involved development of a smart evaporative cooling system for tomato storage, utilizing IoT-based temperature and humidity monitoring to extend freshness by 2-3 days compared to conventional storage methods, reducing post-harvest losses.

- **Extension Experiences**

  Smart Cotton Module Monitoring, Texas, USA: Demonstrated the newly developed AI-based smart monitoring system for round cotton modules to producers and ginners and presented its potential economic benefits for cotton production and ginning operations in different ginning school. Collected end-user feedback and incorporated stakeholder input to improve and adapt the system for practical field and industry applications.

  Seedling Transplantation Aid, South Korea: Demonstrated a newly developed automatic seedling transplantation aid designed to improve safety and reduce physical burden for elderly users. Collected user feedback and incorporated stakeholder input to refine and upgrade the system.

  Greenhouse Climate Management, South Korea: Demonstrated the effectiveness of a newly developed hybrid heater-dehumidifier module integrated with a wireless sensor network (WSN) for monitoring and managing the spatial distribution of temperature and humidity in end users' greenhouses.

  Farm Injury Risk Assessment, Bangladesh: Conducted field surveys to identify agricultural activities associated with higher risks of fatal and serious injuries, assess end-user safety needs, and identify priority areas for immediate intervention. Conducted subsequent research and development to design targeted safety features based on field observations and end-user needs.

- **Teaching Experiences**

  Instructor: *Applied Methods in SmartAg Systems (AOM 5456)* at the University of Florida, Spring 2026. Delivered hands-on instruction in smart agriculture technologies, including SmartAg hardware and software, microcontroller programming, sensing systems, controlled-environment applications, data analytics, and precision agriculture.

  Co-instructor: *Instrumentation in Agricultural Engineering Research (ABE 6031)* (Fall 2025, 2026) and *Power and Machine Design in Agricultural and Biological Engineering (ABE 4171)* (Fall 2026), at the University of Florida. Delivered instruction on theoretical concepts and supervised hands-on projects involving path following, collision avoidance, object tracking, and path planning using Boe-Bot and JetBot platforms, as well as wireless sensor networks, cloud-based computing solutions, robotics applications, automation and control strategies, and decision-support systems. Also, preparing to teach *Applied Control for Automation and Robotics (ABE 6005)* in the upcoming semester, incorporating TurtleBot platforms, Robot Operating System (ROS), and Gazebo for real-world robotic applications and simulation-based exercises.

  Teaching Assistant: *Measurement and Control of Biological Systems and Agricultural Processes (BAEN 370)*, Texas A&M University (Spring 2023, 2024). Contributed to instructional material design; evaluated student performance; and supported instruction in sensor theory, artificial intelligence, and intelligent automation through lectures and hands-on exercises. Guided students in the design of AI-enabled sensing and control systems for biological and agricultural applications, including sensor operation, data acquisition, signal processing, machine learning-based decision-making, control techniques, and integration of automation, robotics, and smart agricultural technologies.

  Lab Instructor: Reuse Water Quality Research and Extension Experiences for Undergraduates (RWQ-REEU), Texas A&M University (RELLIS Campus, 2022). Provided instruction on wastewater treatment processes and supervised student activities in water sample collection and analysis across multiple treatment stages. Covered key parameters including pH, dissolved oxygen, conductivity, *E. coli*, and biochemical oxygen demand (BOD). Trained students in onsite sensing techniques and laboratory analytical methods.

- **Mentoring:** Supervised 5 M.S., 1 Ph.D., and 6 undergraduate students; currently advising 2 M.S. students.

[1] Akash Singh (Master's student - University of Florida) 2026 - Present
Deep learning-assisted detection of *E. coli* using UV-C fluorescence imaging

[2] Sriram Thota (Master's student - University of Florida) 2025 - Present
Development of a plant disease mapping and AI-based diagnostic app

[3] Hemanth Maraboina (Master's student - University of Florida) 2025 - 2026
Development of a plant disease mapping and AI-based diagnostic app

[4] Satya Akash Obellaneni (Master's student - University of Florida) 2024 - 2025
Disease detection in Fruits and leaves using Vision-Transformer and hyperspectral imagery.

[5] Quentin Frederick (PhD student - University of Florida) 2024 - 2025
Citrus disease classification using machine vision on hyperspectral imagery.

[6] Snehit Vaddi (Master's student - University of Florida) 2024 - 2025
Deep learning-assisted detection of *E. coli* using UV-C fluorescence imaging.

[7] Kyu Cheol Han (Master's student - Chungnam National University) 2018 - 2019
Smart IoT framework for greenhouse crop management and cultivation.

[8] Bo-Eun Jang (Master's student - Chungnam National University) 2018 - 2019
Design and development of a tractor-mounted seeding machine.

Undergrad research project groups

[1] Austin Tucek, Jaden Strong, and Porter Miller (Texas A&M University) 2023 - 2024
IoT-Based Solution for Real-Time Tractor Wheel Lag Calculation.

[2] Tamim ul Hasan, Golam Saklain, and Muhaiminul Islam (H.M.D. Science & Tech University) 2015 - 2017
Upgradation of an Evaporative cooling system for storing fresh tomatoes.

- **Graduate Intern:** Shenyang Agricultural University (Fall 2018, Summer 2019)

  Analyzed environmental conditions in Chinese solar greenhouses across summer and winter to identify energy-saving opportunities. Collected continuous data using a wireless sensor network, including temperature, humidity, light intensity, CO_2 concentration, solar radiation, and wind speed/direction.

- **Grant proposal writing**
  - A Smart Cotton Module Monitoring System for Sustainable Cotton Production (Submitted: 2022 Southern-SARE).
  - Assisted in:
    - Machine Vision and Artificial Intelligence for Smart Farm Innovation and Technology Development (Submitted & funded: Ministry of Agriculture, Food and Rural Affairs (MAFRA), Republic of Korea, 2018).
    - Investigation of the Characteristics and Risk Factors Associated with Fatal Farm and Agricultural Injuries in Bangladesh (Submitted & funded: University Grants Commission of Bangladesh, 2015).

Training & Certifications

- Process-Focused Assessment Design: Tools for Making Expectations and Learning Visible in the AI Era 2026
- Center for the Integration of Research, Teaching, and Learning (CIRTL) Associate 2026
- Preparing Future Faculty program- University of Florida 2026
- Academy for Future Faculty Fellow- Texas A&M University 2024
- RecurDyn (Multibody Dynamics Simulation Software) Professional Training 2018

Skills

- **Artificial Intelligence & Machine Learning:** Python, C++, Machine Learning, Deep Learning, CNNs, Vision Transformers (ViTs), TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, AI-based image analysis, computer vision, multimodal data fusion, and model evaluation.
- **Software & Application Development:** Smartphone app development, mobile interfaces, GUI development, AI model integration, and data visualization.
- **Data Analytics & Image Processing:** NumPy, Pandas, SciPy, MATLAB, R, ArcGIS, data preprocessing, feature extraction, PCA, statistical analysis, signal processing, and hyperspectral and multispectral data analysis.
- **Computer Vision & Imaging Technologies:** Machine vision, hyperspectral and multispectral imaging, fluorescence imaging, RGB/RGB-D imaging, image enhancement, and automated vision-based sensing.
- **Engineering Design & Embedded Systems:** ANSYS, SolidWorks, CATIA, RecurDyn, sensor integration, DAQs, NVIDIA Jetson, microcontrollers (Arduino, Raspberry Pi, ESP32, ATmega128), actuator control.
- **Robotics:** Boe-Bot, JetBot, TurtleBot, Clearpath Ridgeback, Rover MITI, Rover MEGA, Kinova.
- **IoT & Smart Agricultural Systems:** Wireless sensor networks, LabVIEW, RFID, GNSS, LoRa, XBee, Wi-Fi communication, cloud platforms (ThingSpeak, Firebase), precision agriculture technologies, and agricultural system modeling.

Awards & Honors

[1]	Distinguished Graduate Student Award for Excellence in Research, Texas A&M University	2025
[2]	Travel Award from BAEN Graduate Student Association, Texas A&M University	2024
[3]	Dr. Brock Faulkner '04 Memorial Scholarship, from Department of BAEN, Texas A&M University	2023
[4]	Travel Award from BAEN Graduate Student Association, Texas A&M University	2023
[5]	Graduate Scholarship from Department of BAEN, Texas A&M University	2023
[6]	Peary Wilemon scholarship awards from National Cotton Ginners Association, USA	2023
[7]	Travel Award from BAEN Graduate Student Association, Texas A&M University	2023
[8]	3rd place for oral presentation in Student Research Week, Texas A&M University	2022
[9]	Graduate Scholarship from Department of BAEN, Texas A&M University	2022
[10]	1st place for oral presentation in Belt Wide Cotton Conferences, San Antonio, TX, USA	2022
[11]	Peary Wilemon scholarship awards from National Cotton Ginners Association, USA	2021
[12]	Graduate Scholarship from Department of BAEN, Texas A&M University	2021
[13]	Best 3rd oral presenter in China-South Korea Joint Academic Conference, Shenyang, China	2019
[14]	Silver prize for oral presentation in Japan China Korea Graduate Student Forum, Tsukuba, Japan	2018
[15]	Best poster presenter in Spring conference on precision agriculture, Sangju, South Korea	2018

Services

- **Guest Editor:**
  - Sensors: Spectroscopy and Sensing Technologies for Smart Agriculture.
  - AgriEngineering: Sensor-based technologies and machine learning in precision agriculture.
- **Peer Reviewer:** ASABE, COMPAG, Real-Time Image Processing, Smart Agriculture, Sensors, Sustainability, Scientific Reports, Applied Sciences, Agronomy, Water, Air, & Soil Pollution.

  Reviewed 24 different journal articles related to the use of engineering in agriculture. The topics included agricultural machine design and optimization, the application of artificial intelligence, machine vision, robotics, convolutional neural networks, and automation and control.
- **Panelist:**
  - [1] Poster Symposium, University of Florida 2025
  - [2] Capstone Project Competition, Texas A&M University 2024
  - [3] Capstone Project Competition, Texas A&M University 2023
  - [4] Student Research Week (SRW), Texas A&M University 2023
  - [5] 3MT presentation, Texas A&M university 2023
  - [6] ASABE conference, Houston, TX 2022
- **Senator:** Graduate & Professional Student Government, Texas A&M University 2021-2022
- **Secretary:** BAEN Graduate Student Association, Texas A&M University 2021-2022
- **Organizing Secretary:** Bangladesh Student Association, Texas A&M University 2020-2022
- **General Secretary:** Engineering Students' Association, Hajee Mohammad Danesh Science & Technology University, Bangladesh 2015-2017
- **Community Service:** Volunteer at UF Organic Community Garden; maintained shared spaces, compost, walkways, and community beds to support sustainable campus agriculture. 2025- Present
- **Professional Memberships:**
  - [1] Sigma Xi, The Scientific Research Honor Society 2026 - Present
  - [2] Society of Photo-Optical Instrumentation Engineers (SPIE) 2024 - Present

[3] American Society of Agricultural and Biological Engineers (ASABE) 2018 - Present
[4] Precision Cotton Research Group 2020 - 2024

Research Projects

[1] Embedded AI techniques for compact optical sensing systems for food safety and plant health applications, funded by USDA ARS. 2024 - Present
[2] Autonomous Robotic Electric Powered 2-wheel Drive Articulated Platform for CEA, funded by USDA ARS. 2024 - Present
[3] Cotton module handling logistics optimization and cover damage reduction, Supported by Cotton Incorporated, USA. 2020 - 2024
[4] Development of high-tech machinery to improve mechanization on upland-field farming, funded by Ministry of Agriculture, Food and Rural Affairs (MAFRA), South Korea. 2017 - 2019
[5] Industrialization of user comfort improvement technologies for smart greenhouse ICT equipment, supported by the Rural Development Administration, South Korea. 2017 - 2019
[6] Development of remote monitoring sensor and control interface mechanical technology for IoT-based agricultural and livestock production for smart farms, funded by the Ministry of Agriculture, Food and Rural Affairs (MAFRA), South Korea. 2017 - 2019
[7] Development of a low-cost smart evaporative cooling system for storing fresh tomatoes, supported by Hajee Mohammad Danesh Science & Technology University research fund, Bangladesh. 2015 - 2017
[8] Investigation of the Characteristics and Risk Factors Associated with Fatal Farm and Agricultural Injuries in Bangladesh. Supported by the University Grants Commission of Bangladesh. 2016 - 2017

Publications

Peer Reviewed

[1] **Z. Iqbal**, T. F. Burks, S. Thota, et al., (2026). Handheld UV-C Fluorescence Imaging with Cascaded Denoising and Deep Learning for *Escherichia coli* Concentration Classification on Diverse Food-Contact Surfaces. *Sensors*. https://doi.org/10.3390/s26216699
[2] Q. Frederick , T.F. Burks, **Z. Iqbal**, et al., (2026). Multiclass Citrus Leaf Disease Inspection with Region Classification and Hyperspectral Imagery. *Journal of the ASABE.* https://doi.org/10.13031/ja.16525
[3] J. Wanjura, J. Ward, W. Porter, L. Fuhrer, B. Hardin, **Z. Iqbal**, and E. Barnes. (2026). RFID-What's in it for me? Crop Protection Network. https://doi.org/10.31274/cpn-20260227-0 (Extension article)
[4] **Z. Iqbal**, T. F. Burks, S. Vaddi, et al., (2026). Detecting *Escherichia coli* on Conventional Food Processing Surfaces Using UV-C Fluorescence Imaging and Deep Learning. *Applied Sciences*. https://doi.org/10.3390/app16020968
[5] S. Vaddi, T.F. Burks, **Z. Iqbal**, et al., (2025). Detecting *Escherichia coli* Contamination on Plant Leaf Surfaces Using UV-C Fluorescence Imaging and Deep Learning. *Plants*. https://doi.org/10.3390/plants14213352
[6] **Z. Iqbal,** R.G. Hardin, E.D. Barnes, et al., (2025). Cover Damage Detection in Round Cotton Modules using Convolutional Neural Networks (CNNs). *Computers and Electronics in Agriculture*. https://doi.org/10.1016/j.compag.2025.111023
[7] **Z. Iqbal,** T. Burks, P. Yadav, et al., (2025). A Robust Framework Combining Hyperspectral Imaging and Machine Learning for Assessing Sudden Death Syndrome (SDS) Severity in Soybean Foliage. *Journal of Biosystems Engineering*. https://doi.org/10.1007/s42853-025-00278-9
[8] M.N. Islam, M.N. Reza, **Z. Iqbal,** et al., (2025). Spatial and Temporal Variability of Environmental Variables in Chinese Solar Greenhouses in the Summer Season. Horticulture. https://doi.org/10.3390/horticulturae11040421
[9] **Z. Iqbal,** M.N. Islam, M.S.N. Kabir, et al., (2023). Comparison of heating modules for suspension-type multipoint temperature variability management in smart greenhouses. *Smart Agricultural Technology*. https://doi.org/10.1016/j.atech.2023.100296
[10] M.N. Reza, M.N. Islam, **Z. Iqbal,** et al., (2023). Spatial, Temporal, and Vertical Variability of Ambient Environmental Conditions in Chinese Solar Greenhouses during Winter, *Applied Sciences*. https://doi.org/10.3390/app13179835
[11] M.N. Islam, **Z. Iqbal,** M. Ali, et al., (2023). Evaluation of a 0.7 kW Suspension-Type Dehumidifier Module in a Closed Chamber and in a Small Greenhouse, *Sustainability*. https://doi.org/10.3390/su15065236
[12] M.N. Islam, **Z. Iqbal,** M. Ali, et al., (2022). Theoretical transmission analysis to optimize Gearbox for a 2.6 kW automatic pepper transplanter, *Journal of Agricultural Engineering*. https://doi.org/10.4081/jae.2022.1254

[13] **Z. Iqbal,** M.N. Islam, M. Ali, et al., (2022). Theoretical Overturning Analysis of a 2.6-kW Two-Row Walking-Type Automatic Pepper Transplanter. *Journal of Biosystems Engineering*. https://doi.org/10.1007/s42853-022-00129-x
[14] M.N. Islam, **Z. Iqbal**, M. Ali, et al., (2021). Evaluation of pepper seedling growth according to the growing period and tray for automatic transplanting, *Korean Journal of Agricultural Science*. https://doi.org/10.7744/kjoas.20210080
[15] **Z. Iqbal,** M.N. Islam, M.S.N. Kabir, et al., (2021). Kinematic analysis of a hopper-type dibbling mechanism for a 2.6 kW two-row pepper transplanter, *Journal of Mechanical Science and Technology*. https://doi.org/10.1007/s12206-021-0531-2
[16] M.N. Islam, **Z. Iqbal**, M. Chowdhury, et al., (2021). Stress and fatigue analysis of picking device gears for a 2.6 kW automatic pepper transplanter, *Applied Sciences*. https://doi.org/10.3390/app11052241
[17] **Z. Iqbal,** M.N. Islam, M. Chowdhury, et al., (2021). Working speed analysis of the gear-driven dibbling mechanism of a 2.6 kW walking-type automatic pepper transplanter, *Machines*. https://doi.org/10.3390/machines9010006
[18] M.N. Islam, **Z. Iqbal,** M. Ali, et al., (2020). Kinematic analysis of a clamp-type picking device for an automatic pepper transplanter, *Agriculture*. https://doi.org/10.3390/agriculture10120627
[19] M. Chowdhury, M.N. Islam, **Z. Iqbal,** et al., (2020). Analysis of overturning and vibration during field operation of a tractor-mounted 4-row radish collector toward ensuring user safety, *Machines*. https://doi.org/10.3390/machines8040077
[20] M.N. Islam, **Z. Iqbal,** M. Ali, et al., (2020). Performance Evaluation of a Suspension-Type Dehumidifier with a Heating Module for Smart Greenhouses, *Journal of Biosystems Engineering*. https://doi.org/10.1007/s42853-020-00055-w
[21] **Z. Iqbal,** N. Islam, B.E. Jang, M. Ali, et al., (2019). Monitoring the Operating Status of an Automatic Harmful Fly Collector for Smart Greenhouses, *Journal of Biosystems Engineering*. https://doi.org/10.1007/s42853-019-00036-8
[22] M.N. Islam, **Z. Iqbal,** M.S.N. Kabir, et al., (2019). Performance Evaluation of Trenchless Subsurface Drainage Piping Machine, *Journal of Biosystems Engineering*. https://doi.org/10.1007/s42853-019-00032-y

Conference Proceedings

[1] **Z. Iqbal**, T.F. Burks, S. Thota, et al., (2026). Vision Transformer-Based Fusion of Hyperspectral Reflectance and Fluorescence Imaging for Citrus Leaf Disease Classification. In Proc. of ASABEAIM2026, Indianapolis, IN.
[2] **Z. Iqbal**, T.F. Burks, S. Obellaneni, et al., (2026). Integration of RGB and hyperspectral reflectance imagery using vision transformer for citrus leaf disease classification. In Proc. of SPIE Defense + Security, National Harbor, MD. https://doi.org/10.1117/12.3095060
[3] T.F. Burks, **Z. Iqbal**, Sriram Thota, et al., (2026). AI-based detection of *E. coli* concentration levels on diverse food-contact surfaces using UV-C fluorescence-based CSI-D+ handheld and Yolov11 deep learning. In Proc. of SPIE Defense + Security, National Harbor, MD. https://doi.org/10.1117/12.3095051
[4] Q. Frederick , T.F. Burks, **Z. Iqbal**, et al., (2026). A multiclass classification approach for citrus leaf disease detection using small regions of hyperspectral imagery and CNNs. In Proc. of SPIE Defense + Security, National Harbor, MD. https://doi.org/10.1117/12.3094086
[5] T.F. Burks, **Z. Iqbal**, Akash Singh, et al., (2026). Enhanced *E. coli* concentration level detection on multiple produce surfaces using CSI-D+ UV-C fluorescence imaging and custom deep learning architecture. In Proc. of SPIE Defense + Security, National Harbor, MD. https://doi.org/10.1117/12.3095024
[6] **Z. Iqbal**, T.F. Burks, P.K. Yadav, et al., (2025). Hyperspectral imaging and machine learning for soybean SDS severity classification. In Proc. of SPIE Defense + Commercial Sensing, Orlando, FL. https://doi.org/10.1117/12.3054913
[7] Q. Frederick , T.F. Burks, **Z. Iqbal**, et al., (2025). Multiclass citrus leaf disease inspection with region classification and hyperspectral imagery. In Proc. of SPIE Defense + Commercial Sensing, Orlando, FL. https://doi.org/10.1117/12.3054678
[8] T.F. Burks, S. Vaddi, **Z. Iqbal**, et al., (2025). Improved E. coli concentration levels detection on citrus leaf using CSI-D+ UV-C fluorescence imaging and custom deep learning using YOLOv11 architecture. https://doi.org/10.1117/12.3054911
[9] Q. Frederick , T.F. Burks, **Z. Iqbal**, et al., (2025). Investigating feature types for automated multiclass citrus peel disease detection. In Proc. of SPIE Defense + Commercial Sensing, Orlando, FL. https://doi.org/10.1117/12.3054673
[10] T.F. Burks, S. Vaddi, **Z. Iqbal**, et al., (2025). Detecting E. coli concentration levels on various surfaces using UV-C fluorescence imaging and YOLO11 deep learning. https://doi.org/10.1117/12.3054912
[11] **Z. Iqbal**, R.G. Hardin, J.K. Ward, et al., (2023). Development of a Moisture and Temperature Monitoring System for Round Cotton Modules. In proc. of 2023 Beltwide Cotton Conferences. New Orleans, LA. https://www.cotton.org/beltwide/proceedings/

[12] **Z. Iqbal**, R.G. Hardin, J.K. Ward, et al., (2022). Round Modules: Handling Logistics and Cover Damage, 3rd Year. In proc. of 2022 ASABE Annual International Meeting. Houston, TX. https://doi.org/10.13031/aim.202200300

[13] **Z. Iqbal**, R.G. Hardin, J.K. Ward, et al., (2022). A CNN-based Approach to Detect Cover Damage of Round Cotton Modules. In proc. of 2022 Beltwide Cotton Conferences. SanAntonio, TX. https://www.cotton.org/beltwide/proceedings/2005-2022/index.htm

[14] **Z. Iqbal**, R.G. Hardin, J.K. Ward, et al., (2021). Round modules: Handling logistics and cover damage, 2nd year. In proc. of 2021 Beltwide Cotton Conferences. https://www.cotton.org/beltwide/proceedings/2005-2022/index.htm

Manuscripts Under Review

[1] **Z. Iqbal**, T.F. Burks, et al. Vision Transformer-Based Multimodal Fusion of RGB and Hyperspectral Imagery for Citrus Leaf Disease Classification. *Computers and Electronics in Agriculture.*

[2] **Z. Iqbal**, T.F. Burks, et al. Comparative Evaluation of Image Denoising Methods for Deep Learning-Based *Escherichia coli* Concentration Level Detection Using UV-C Fluorescence Imaging. *Smart Ag. Technology.*

[3] S. Das,S. Kaur, **Z. Iqbal**, *et al*. Development of an IoT-Based Smart Irrigation System Integrating Crop Water Stress Index (CWSI) for Irrigation Scheduling in Maize-Wheat Cropping Systems. *Earth*

Manuscripts in Progress

[1] **M.Z. Iqbal**, T.F. Burks, et al. Vision Transformer-Based Multimodal Fusion of Hyperspectral Reflectance and Fluorescence Images for Citrus Leaf Disease Classification. *International Journal of Agricultural & Biological Engineering.*

[2] **Z. Iqbal**, R.G. Hardin, E.D. Barnes, et al. A Smart Moisture Monitoring System for Round Cotton Modules. *Journal of the ASABE.*

[3] **Z. Iqbal**, R.G. Hardin, R. Robinson, et al. Smart Cotton Module Tracking and Monitoring System (SCMTM 2.0): An Upgraded SCMTM With Comprehensive Logistics and Economic Analysis. *Journal of Cotton Science.*

Invited Talks

[1] *Toward Autonomous Agriculture: Sensing, Robotics, and AI for Next-Generation Farming* at North Carolina State University, 2026.

[2] *Emerging Research Trends in Smart Agricultural Machinery and Precision Technologies* at Hajee Mohammad Danesh Science and Technology University, Bangladesh, 2025.

Conferences

Oral Presentations

[1] Z. Iqbal*, T.F. Burks, S. Thota, et al., (2026). Vision Transformer-Based Fusion of Hyperspectral Reflectance and Fluorescence Imaging for Citrus Leaf Disease Classification. In Proc. of ASABEAIM2026, Indianapolis, IN.

[2] Z. Iqbal*, T.F. Burks, S. Obellaneni (2026). Integration of vision transformer and hyperspectral reflectance imaging for citrus leaf disease classification. Conference on Sensing for Agriculture and Food Quality and Safety XVIII, part of SPIE Defense + Security. National Harbor, MD.

[3] T.F. Burks, Z. Iqbal*, S. Vaddi (2026). Enhanced *E. coli* concentration level detection on multiple produce surfaces using CSI-D+ UV-C fluorescence imaging and custom deep learning architecture. Conference on Sensing for Agriculture and Food Quality and Safety XVIII, part of SPIE Defense + Security. National Harbor, MD.

[4] T.F. Burks, Z. Iqbal*, S. Vaddi (2026). AI-based detection of *E. coli* concentration levels on diverse food-contact surfaces using UV-C fluorescence-based CSI-D+ Handheld and Yolov11 deep learning. Conference on Sensing for Agriculture and Food Quality and Safety XVIII, part of SPIE Defense + Security. National Harbor, MD.

[5] Z. Iqbal, T.F. Burks, et al. (2025). Hyperspectral imaging and machine learning for soybean SDS severity classification. Presented at the Sensing for Agriculture and Food Quality and Safety XVII, Orlando, FL, USA.

[6] T.F. Burks, S. Vaddi, Z. Iqbal, et al. (2025). Detecting *E. coli* concentration levels on various surfaces using UV-C fluorescence imaging and YOLO11 deep learning. Presented at the Sensing for Agriculture and Food Quality and Safety XVII, Orlando, FL, USA.

[7] Z. Iqbal, R.G. Hardin, J.K. Ward, et al. (2024). Smart Moisture Monitoring for Round Cotton Modules: Development and Testing. Presented at the 2024 American Society of Agricultural and Biological Engineers Annual International meeting, Anaheim, CA, USA.

[8] Z. Iqbal, R.G. Hardin, J.K. Ward, et al. (2024). Strategic Implementation of Convolutional Neural Networks for Simplified Wrap Damage Detection in Round Cotton Modules. Presented at the 2024 Beltwide Cotton Conference, Fort Worth, TX, USA.

[9] Z. Iqbal, R.G. Hardin, J.K. Ward, et al. (2023). A Smart Monitoring System for Quality Cotton Production. Presented at the 2023 American Society of Agricultural and Biological Engineers Annual International meeting, Omaha, NE, USA.

[10] Z. Iqbal, R.G. Hardin, J.K. Ward, et al. (2023). Development of a moisture and temperature monitoring system for round cotton modules. Presented at the 2023 Beltwide Cotton Conference, New Orleans, LA, USA.

[11] Z. Iqbal, R.G. Hardin (2022). A CNN-based Approach to Detect Cover Damage of Round Cotton Modules. Presented in Texas A&M University Student Research Week, College Station, TX, USA.

[12] Z. Iqbal, R.G. Hardin, J.K. Ward, et al. (2022). Round Modules: Handling Logistics and Cover Damage, 3rd Year. Presented at the 2022 American Society of Agricultural and Biological Engineers Annual International meeting, Houston, TX, USA.

[13] Z. Iqbal, R.G. Hardin, J.K. Ward, et al. (2022). Round Modules Monitoring System: A Smart Solution for Quality Cotton Production. Presented at the 3rd Annual Paper Meet of Agricultural Engineering Division of Institution of Engineers. Dhaka, Bangladesh.

[14] Z. Iqbal, R.G. Hardin, J.K. Ward, et al. (2022). A CNN-Based Approach to Detect Cover Damage of Round Cotton Modules. Presented at the 2022 Beltwide Cotton Conference, San Antonio, TX, USA.

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Poster Presentations

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